AWS EC2 + React Frontend + FastAPI Backend

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1. AWS EC2 Setup
2. Backend (FastAPI) Deployment
3. Frontend (React + Vite) Deployment
4. Environment Configuration
5. CORS Configuration
6. Common Issues & Fixes
7. Final Checklist

1.1 — Launch EC2 Instance

- **AMI**: Ubuntu 22.04 LTS
- **Instance Type**: `t2.micro` (free tier) or `t3.small` (production)
- **Storage**: 20 GB gp2
- **Key Pair**: Create and download `.pem` file securely

1.2 — Security Group Rules (CRITICAL)

Type	Protocol	Port	Source
SSH	TCP	22	Your IP
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0
Custom TCP	TCP	**8000**	0.0.0.0/0
Custom TCP	TCP	**5173**	Your IP (dev only)

> ˆ **Port 8000 must be open** for FastAPI to be externally accessible.

1.3 — Connect to EC2

bash
chmod 400 your-key.pemssh -i your-key.pem ubuntu@<YOUR_EC2_PUBLIC_IP>

2.1 — Update System & Install Python

bash
sudo apt update && sudo apt upgrade -ysudo apt install python3 python3-pip python3-venv git -y

2.2 — Clone Your Project

bash
cd ~git clone <YOUR_REPO_URL>cd KashiKartCiCd/backend

2.3 — Create Python Virtual Environment

bash
python3 -m venv venvsource venv/bin/activate

2.4 — Install Dependencies

bash
pip install -r requirements.txt

2.5 — Configure Environment Variables

bash
nano .env

Add your backend variables:

DATABASE_URL=your_database_urlSECRET_KEY=your_secret_keyDEBUG=Fa
lse

2.6 — Test FastAPI Runs

bash
uvicorn main:app --host 0.0.0.0 --port 8000

Visit: `http://<YOUR_EC2_PUBLIC_IP>:8000/docs` — you should see Swagger UI.

2.7 — Run FastAPI as Background Service (Systemd)

bash
sudo nano /etc/systemd/system/kashikart.service

Paste this:

ini
[Unit]Description=KashiKart FastAPI
BackendAfter=network.target[Service]User=ubuntuWorkingDirectory=/home/ubun
tu/KashiKartCiCd/backendEnvironment="PATH=/home/ubuntu/KashiKartCiCd
/backend/venv/bin"ExecStart=/home/ubuntu/KashiKartCiCd/backend/venv/bin
/uvicorn main:app --host 0.0.0.0 --port
8000Restart=always[Install]WantedBy=multi-user.target

Enable and start:

bash
sudo systemctl daemon-reloadsudo systemctl enable kashikartsudo systemctl start kashikartsudo systemctl status kashikart

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3.1 — Install NVM (Node Version Manager)

bash
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash

Load NVM immediately (without restarting terminal):

bash
export NVM_DIR="$HOME/.nvm"[-s "$NVM_DIR/nvm.sh"] && \. "$NVM_DIR/nvm.sh"

3.2 — Install Node.js LTS

bash
nvm install --ltsnvm use --ltsnode -v # Should show v20.x.x or similarnpm -v

3.3 — Navigate to Frontend Directory

bash
cd ~/Claude-DevOps-Workspace/KashiKartCiCd/frontend

3.4 — Clean Install Dependencies

bash
rm -rf node_modules package-lock.jsonnpm install

4.1 — Create `.env` File in Frontend Root

bash
Make sure you're in the frontend directorycd ~/Claude-DevOps-Workspace/KashiKartCiCd/frontend# Create .env with correct value (run this in terminal, NOT paste into file)echo "VITE_API_BASE_URL=http://<YOUR_EC2_PUBLIC_IP>:8000" > .env

4.2 — Verify `.env` Content

bash
cat .env

Expected output (exactly one clean line):

VITE_API_BASE_URL=http://13.205.245.107:8000

> ˆ **Common mistake**: Do NOT paste the `echo` command text inside the `.env` file. Run it in terminal.

`vite.config.js`

If `vite.config.js` is missing, create it:

bash
cat > vite.config.js << 'EOF'import { defineConfig } from 'vite'import react from '@vitejs/plugin-react'export default defineConfig({ plugins: [react({ jsxRuntime: 'automatic' })], server: { port: 5173, host: true }})EOF

6.1 — Fix All JSX Files Missing `import React`

bash
for f in $(grep -rL "import React" ./src --include="*.jsx"); do sed -i " '1s/^/import React from '''''react''''';\n/' "$f" echo "Fixed: $f"done``*

> **Note**: With `jsxRuntime: 'automatic'` in `vite.config.js`, this step becomes optional for new files.

The FastAPI backend must allow requests from your frontend origin.

7.1 — View Current CORS Config

bash
On EC2:grep -n "CORS\|allow_origins\|CORSMiddleware" ~/KashiKartCiCd/backend/main.py

7.2 — Update `allow_origins` in `main.py`

python
from fastapi.middleware.cors import CORSMiddlewareapp.add_middleware(CORSMiddleware, allow_origins=["http://localhost:5173", "http://<YOUR_FRONTEND_DOMAIN>",

"https://<YOUR_FRONTEND_DOMAIN>"], allow_credentials=True, allow_methods=["*"], allow_headers=["*"],)

7.3 — Restart Backend After CORS Change

bash
sudo systemctl restart kashikartsudo systemctl status kashikart

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bash
cd ~/Claude-DevOps-Workspace/KashiKartCiCd/frontendnpm run dev

Visit: `http://localhost:5173`

9.1 — Build the Frontend

bash
npm run build

This generates a `dist/` folder.

9.2 — Serve with Nginx on EC2 (Production)

Install Nginx on EC2:

bash
sudo apt install nginx -y

Copy build files:

bash
sudo cp -r dist/* /var/www/html/``*

Configure Nginx:

bash
sudo nano /etc/nginx/sites-available/kashikart
nginx
server { listen 80; server_name <YOUR_EC2_PUBLIC_IP>; root /var/www/html; index index.html; location / { try_files $uri $uri/ /index.html; } location /api/ { proxy_pass http://127.0.0.1:8000/; proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr; }}

Enable and restart Nginx:

bash
sudo ln -s /etc/nginx/sites-available/kashikart /etc/nginx/sites-enabled/sudo nginx -tsudo systemctl restart nginx

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Error	Fix
Port 8000 not accessible externally	Open port 8000 in EC2 Security Group
Error [ERR_MODULE_NOT_FOUND]: vite	Run rm -rf node_modules && npm install
nvm: command not found	Run export NVM_DIR="$HOME/.nvm" && \. "$NVM_DIR/nvm.sh"
.env has echo text literally	Run echo "VITE_API_BASE_URL=..." > .env in terminal
ReferenceError: React is not defined	Add import React from 'react' to JSX files or set jsxRuntime: 'automatic'
vite.config.js not found	Create it manually with the template in Part 5
Layout.jsx not in src/components/	Actual path is src/layout/Layout.jsx
Dashboard shows "Something went wrong"	Fix .env file and restart dev server

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FINAL DEPLOYMENT CHECKLIST

□ EC2 instance launched with correct security group (port 8000, 80, 443 open)□ SSH access confirmed□ Backend dependencies installed in virtualenv□ Backend .env configured□ FastAPI running and accessible at http://<EC2_IP>:8000/docs□ FastAPI set up as systemd service (auto-restart on reboot)□ CORS updated to include frontend origin□ Node.js LTS installed via nvm□ Frontend node_modules cleanly installed□ Frontend .env contains VITE_API_BASE_URL=http://<EC2_IP>:8000□ vite.config.js exists with jsxRuntime: 'automatic'□ All JSX files have import React (or automatic runtime enabled)□ npm run dev works locally□ npm run build generates dist/ folder□ Nginx configured to serve frontend and proxy /api to FastAPI□ Full app accessible at http://<EC2_IP>